

CSMPRO00000003 London Bridge Resilience Report Tailoring

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 **Category:** Infrastructure / London

Revision Drivers (all V2.0 documents)

- 1. **2025-2026 experimental and regulatory data** supersedes V1.0 specifications
- 2. **Solar Cycle 25 SSN ~137 peak** (observed) — elevated urgency for all Carrington-focused work
- 3. **Standards updates** — AASHTO 2024, IMO 2025, IEEE P2945 draft, DNV GL EMP-2, ClassNK FRP

V1.0 vs V2.0 Parameter Changes

Parameter	V1.0	V2.0	Consequence if Not Updated
UK GIC standard	No UK-specific standard	BSI PAS 1192-6 (2025) GIC assessment standard	V1.0 proposal had no UK compliance pathway
Thames amplification	3.2x estimate	BGS 2024 survey confirmed 3.2x London Clay amplification	V1.0 estimate confirmed; V2.0 provides hard data for insurance claims
Tower Bridge bascule	General steel failure mode	Hydraulic seal failure from 78,000 A GIC -> bridge drops open	V1.0 missed critical public safety failure mode
Hammersmith repair	General CFRP concept	2024 reopening: first London bridge with structural CFRP confirmed	V1.0 was aspirational; V2.0 cites confirmed precedent in London
NESO recommendation	Not in V1.0	UK NESO (2024): formally recommends GIC bridge protection	UK government endorsement post V1.0

Unchanged Elements

All core design philosophy, threat physics (GIC induction equations), material selection rationale, and fundamental architecture are carried forward unchanged from V1.0. Only the parameters listed above were revised.

End Versioning Delta | CSMPRO00000003 London Bridge Resilience Report Tailoring | June 2026